Feasibility Assessment of Denial-of-Service Attacks by Analyzing SOME/IP-SD State Transition Models

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Abstract—In recent years, automotive software have gradually become larger and more complex, thereby increasing the demand for over-the-air software updates after vehicle delivery. Automotive open system architecture (AUTOSAR) specifies scalable service-oriented middleware over IP (SOME/IP) and SOME/IP-service discovery (SOME/IP-SD) as service-oriented communication middleware. These are not designed to encrypt or authenticate. It has a problem that a spoofing attack like man-in-the-middle can be easily initiated. Intrusion detection systems (IDSs) have been proposed for SOME/IP; however, the IDS may not be able to detect an attack when normal device transmission is blocked and only the attacker transmits data. In this study, we demonstrated the feasibility of denial-of-service attacks that disable service provisioning and usage through an exhaustive analysis of the state-transition model of SOME/IP-SD. An attack on a disabled service provision is a threat that has not been reported in previous studies, and therefore, we evaluated attack feasibility by analyzing the specifications of software components and demonstrating the attack on a real device using a prototype implementation. Implementations used in the evaluation did not consider the specifications utilized in the attack, rendering the proposed attack unfeasible. Additional evaluations conducted in an environment with additional implementations of the specifications used by attacks showed that the proposed attacks were feasible when the implementations were compliant with SOME/IP-SD specifications. Based on the evaluation results, an example of an effective countermeasure against such attacks is provided.

Index Terms—Automotive Ethernet, DoS, SOME/IP, SOME/IP-SD

I. Introduction

In recent years, the emergence of advanced driver-assistance systems such as lane keeping assist and automatic braking has made software used in vehicles larger and more complex, increasing the demand for software and hardware updates, functionality additions, and upgrades through over-the-air and plug-and-play mechanisms after vehicle delivery. Service-oriented architectures have gained considerable attention to address these problems. Automotive open system architecture (AUTOSAR), a global development partnership for the automotive industry has developed scalable service-oriented middleware over IP (SOME/IP)[1] and SOME/IP-service discovery (SOME/IP-SD)[2]. The former is an application-layer protocol for the TCP/IP stack and does not have encryption and authentication mechanisms. A previous study [3] analyzed the SOME/IP specification and reported that a man-in-the-middle attack, which spoofs a server providing a service, can be initiated. Another study [4] discussed the use of IPSec and TLS/DTLS in SOME/IP and concluded that there are challenges in satisfying the strict performance requirements [5][6] for the communication establishment time and communication delay in automotive systems. One other study [7] proposed an intrusion detection system (IDS) for detecting spoofing attacks using SOME/IP; however, the IDS could not detect messages sent by only the attacker because normal devices do not send messages. In a controller area network (CAN), which is used widely as a network communication protocol for in-vehicle systems, the feasibility of an attack that sends spoofed messages after a denial-of-service (DoS) attack disables normal device transmission and reception has been reported [8]. The aforementioned DoS attack [8] may not be applicable to SOME/IP because it is based on an error-control specification of the CAN protocol. However, analyzing the possibility of DoS attacks that block normal device transmissions is necessary for securing communication channels in SOME/IP.

This paper presents a possible DoS attack against network communication using SOME/IP that blocks the provision of services by a server and the use of services by a client. In addition, we evaluate the attack by analyzing the specifications of software components and demonstrating the attack on real devices.

The contributions of this study are as follows:

- We show that an attacker on the same network can realize an attack to disable service provision by a server and service usage by a client. The conditions to realize such an attack are determined through an exhaustive analysis of the SOME/IP-SD state transition model. To the best of our knowledge, this is the first study that demonstrates the threat posed by an attack that denies service provision by a server.
- We analyze the AUTOSAR service discovery (SD)[9] specification to evaluate the feasibility of the attack to deny service provisioning by the server and conduct a real device evaluation using a prototype implementation of SOME/IP, vsomeip [10]. The evaluation

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results indicate that some protocol specifications are not implemented, and therefore, the proposed DoS attack on the server is not feasible. Additional experiments demonstrate that if vsomeip is implemented according to the protocol specifications, the DoS attack proposed in this study is feasible.
- Based on the analysis results and evaluation of the SOME/IP-SD state transition model conducted in this study, an example of an effective countermeasure against the DoS attack in SOME/IP is demonstrated.

The remainder of this paper is organized as follows: Section II outlines the requirements for in-vehicle networks, details the characteristics of SOME/IP and SOME/IP-SD protocols used in in-vehicle networks, examines previous attacks on in-vehicle networks, proposes countermeasures to address these security concerns, and discusses the associated challenges. Section III presents intrusion-detection and communication-protection mechanisms used in the proposed method. Section IV evaluates the feasibility of the DoS attack presented in Section III by analyzing the AUTOSAR SD specification and evaluating a real device using vsomeip [10], which is a prototype implementation of SOME/IP, and discusses countermeasures. Finally, Section V presents the conclusions and directions for future research.

II. Related Work

A. SOME/IP

SOME/IP is a specification of service-oriented automotive middleware used for controlling communication messages [1]. SOME/IP supports the following functions when used in combination with SOME/IP-SD.

- Serialization
  Conversion to a communication format.
- Remote procedure call and messaging
  Remote function calls and other messaging.
- Event notifications
  Notification from the service application (server) to the client.
- SD
  Dynamic SD and access control.
- Publish/subscribe (Pub/Sub)
  Dynamically configure communication between the server and client.
- Segmentation of UDP messages
  SOME/IP messages transmitted via UDP without fragmentation.

SOME/IP does not have its own encryption or authentication mechanisms.

B. SOME/IP-SD

SOME/IP-SD is a middleware specification that discovers available services in the system (SD) and manages event messages sent (Pub/Sub) on SOME/IP [2]. It uses messages listed in TABLE I for SD. These messages include TTL [units: second], which indicates the validity period. An OFFER SERVICE with TTL = 0 is called a STOP OFFER SERVICE and used to indicate that service provision has stopped. The server supports either unicast or multicast transmissions to the client that sent the FIND SERVICE. However, multicast transmission is optional and enabled only when at least half of the transmission cycle has elapsed since the last OFFER SERVICE was transmitted.

TABLE I
Service discovery messages of SOME/IP-SD

Type	Description
FIND SERVICE	Send by client (Multicast) Notification of search for service provide
OFFER SERVICE	Send by Server (Unicast or Multicast) Notification that service is currently available
STOP OFFER SERVICE	Send by server (Multicast) Notification of stop of service provision

The SD function shares information on the available services between the server and client in three phases: initial wait, repetition, and main phases. The client recognizes that the target service is available by receiving the OFFER SERVICE and can then begin using the service.

1) Initial wait phase
   The server and the client wait for a fixed period of time; however, the client immediately moves to the main phase when it receives an OFFER SERVICE.
2) Repetition phase
   Both the server and client send OFFER SERVICE / FIND SERVICE with increasing transmission intervals and transition to the main phase after a fixed number of transmissions are completed. The client moves to the main phase immediately when it receives an OFFER SERVICE message.
3) Main phase
   The server transmits an OFFER SERVICE at regular time intervals, and the client does not transmit a FIND SERVICE.

The server has a Not Ready state, and the client has a Not Requested / Requested_but_not_Ready state. In these states, the service cannot be provided or is not used; they are triggered by application-specific internal states in the server or client, such as device shutdown. A list of the transition conditions for these states is presented in Fig.1 and 2, which are obtained from the figure in the specification [2].

C. In-vehicle Network Attacks and Security

The CAN is a widely used network communication protocol for in-vehicle systems; it is vulnerable to spoofing attacks because the source terminal cannot be identified. CAN messages can be injected to the CAN bus through the OBD-II diagnostic port, which can cause an in-vehicle device to operate [11], [12]. SOME/IP can identify the

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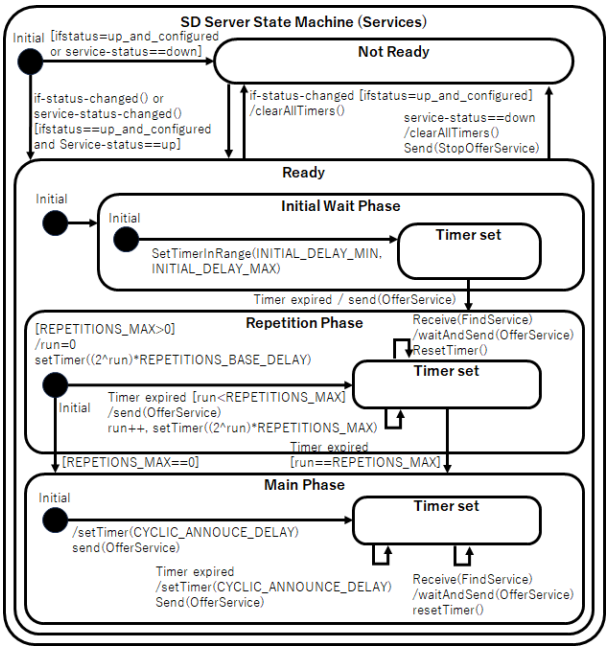

Fig. 1. Server state transition model

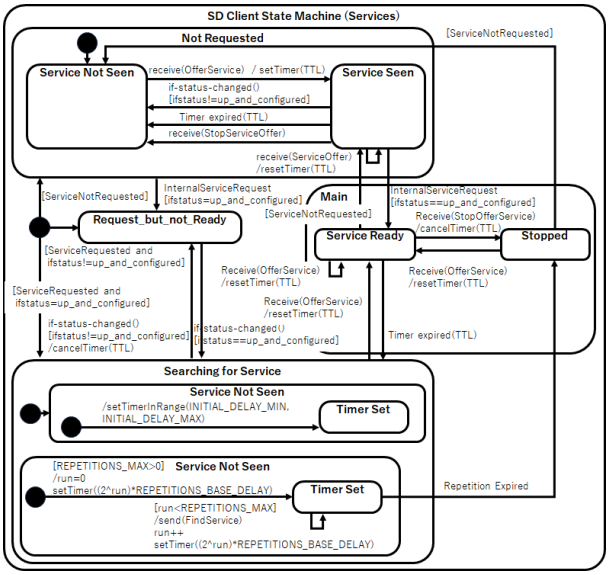

Fig. 2. Client state transition model

source device; however, it does not have a standard authentication mechanism, rendering it possible to spoof the device information. This attack pattern is characterized by mixed communication between normal devices and spoofing attacks. IDSs proposed in previous studies [7],[13] could detect this attack pattern, DoS attacks (called bus-off attacks [14]), and spoofing attacks that used bus-off attacks [8] in CAN. This attack pattern is characterized only via communication with the spoofing attack and reported to be difficult to detect because of monitoring communication cycles [8]. DoS attacks that have the same characteristics as bus-off attacks are yet to be reported for SOME/IP. Further, ARP spoofing [15] can be used to prevent unicast messages sent by the server from reaching the client; however, it is not effective because the server sends an OFFER SERVICE by multicast. Nevertheless, it is important to evaluate the feasibility of the DoS attacks that have characteristics similar to bus-off attacks in SOME/IP for correctly understanding the threats in SOME/IP.

D. SOME/IP Attacks

A previous study [6] analyzed SOME/IP specification and reported that a man-in-the-middle attack that spoofed a server could be employed. Two man-in-the-middle attack methods were reported. The first copies the OFFER SERVICE sent by the server and sends it to the client, thereby causing the client to use services provided by the attacker. This attack is called a copycat attack. In the second method, an attacker spoofs the server and transmits a STOP OFFER SERVICE to the client while the client is using the service, thereby stopping the use of the service and transmitting an OFFER SERVICE to the client to cause the client to use the service provided by the attacker. This attack is called a de-association attack. In these attacks, the server transmission is not disabled, and the client receives the message transmitted by both the server and the attacker. Therefore, there is a high probability that an IDS can detect an attack by monitoring its cycle. Although not reported in previous studies, if a DoS attack that disables the server is feasible, it could pose a greater threat than spoofing attacks reported in previous studies.

In this study, we report the feasibility of a DoS attack that disables service provision by the server and service usage by the client.

III. Attack Method

We demonstrate the possibility of the DoS attacks and the conditions under which such attacks are possible by comprehensively analyzing the state transition model of SOME/IP-SD. We present the characteristics of the attacker and analysis policy of the proposed method, propose attacks to deny service offers by analyzing the state transition model of a server, and deny service use by analyzing the client's state transition model.

A. Attack Model

We assume that some terminals are hijacked or attack devices are connected. We assume that an attacker satisfies the following characteristics:

- Connected on the same network as a normal device (server, client).
- Can send messages that do not conform to SOME/IP and SOME/IP-SD specifications.
- No access (read/write) to the internal information of devices other than that of the attacker.

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- Cannot change the physical connection of the network, including that of the attacker.

This study aims to verify the feasibility of an attack that blocks normal devices from transmitting messages and transmits spoofed messages. Further, we evaluated the feasibility of a DoS attack that prevents only the target terminal from transmitting messages, rather than a DoS attack that increases network bandwidth usage and blocks message transmission. We analyzed only SOME/IP-SD and evaluated the feasibility of the following two attacks.

- Blocking the sending of the OFFER SERVICE by the server
  If this is possible, a spoofing attack in which only the attacker provides the service is possible, thereby rendering detection by the IDS difficult.
- Transitioning client to a state other than Service Ready
  The client stops using the service if this can be achieved, thereby rendering it difficult for the IDS to detect an attack in which an attacker spoofs the client.

We analyzed the state transitions in SOME/IP-SD to comprehensively analyze and verify the feasibility of DoS attacks on SOME/IP-SD. However, the analysis does not cover the Not Ready state of the server or the Not Requested / Requested_but_not_Ready state of the client, which are triggered by the internal state of the device, because the attacker cannot access the internal state of the target terminal.

B. Deny Service Offer

TABLE II lists the state-transition table of the server and results of the analysis of the possibility of an attacker's involvement in each state transition. We focus on state transitions 5 and 8, in which an attacker can participate, and we evaluate the possibility of preventing the server from sending an OFFER SERVICE. Cond. A) and B) are satisfied, and the communication sequence of the service search is shown in Fig.3. We define the behavior of an attacker for achieving this communication as a timer-reset attack.

Cond. A) Attacker transmits the FIND SERVICE at an interval less than half of the Server's internal Timer (i.e., the transmission cycle of OFFER SERVICE).

Cond. B) Attacker transmits the OFFER SERVICE at intervals of less than TTL using an Instance ID that is different from that of the server

In this communication sequence, although the server has not stopped sending the OFFER SERVICE, it is yet to reach the client, except for the first transmission. After the TTL time included in the OFFER SERVICE elapses, the client cannot use the target service, which indicates that a DoS attack has occurred. This attack is achieved by an attacker performing the following two steps to satisfy

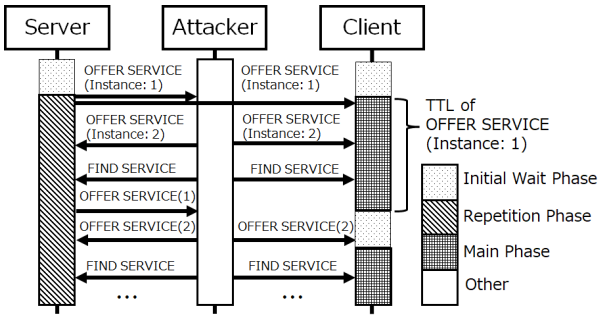

Fig. 3. Timer-Reset attack

the condition that the server does not send an OFFER SERVICE.

- Transmitting the FIND SERVICE resets the timer inside the server and blocks the transmission of OFFER SERVICE by multicast.
- Transmitting the OFFER SERVICE blocks the client from transmitting the FIND SERVICE. As a result, it blocks the server from transmitting the OFFER SERVICE to a normal client by unicast when the server receives a FIND SERVICE.

In contrast to the copycat attack reported in a previous study [3], the timer-reset attack enables the client to use only the services provided by the attacker because the client no longer receives the OFFER SERVICE transmitted by the server.

C. Deny Service Use

TABLE III summarizes the state transition table of the client and the analysis results of the possibility of an attacker's involvement in each state transition. We focus on state transitions 3, 6, 7, 9, 10, 11, and 12, in which an attacker can be involved, and we evaluate the possibility of moving the client to a state besides the Service Ready state. State transitions 3, 6, 9, and 12 cannot block the client from using the service because the destination state is Service Ready. Next, we analyze state transition 11. The communication sequence of SD when Cond. C) and D) are satisfied is shown in Fig.4. We define the behavior of the attacker that enables this communication to proceed as a TTL-override attack.

Cond. C) The attacker sends an OFFER SERVICE to the client immediately after receiving an OFFER SERVICE sent by the server, with the TTL rewritten to 1 s.

Cond. D) The TTL and transmission interval of the OFFER SERVICE sent by the server is greater than 1 s.

In the TTL-override attack, the client enters the initial wait phase 1 s after the attack, and the service is unavailable until the OFFER SERVICE sent by the server is received. The TTL-override attack can be executed for a time longer than 1 s; however, it does not provide any

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TABLE II
Server state transition table and attacker involvement analysis results

No	Old state	New state	Triger	Action	Attacker influence potential
1	-	Timer set of Initial Wait Phase	Imediate	Set Timer in range (min, max)	Not exist: Triger is Immediate
2	Timer set of Initial Wait Phase	Timer set of Repetition Phase	Timer expired	Send OFFER SERVICE, Set Timer, run = 0	Not exist: Timer is internal Variables
3	Timer set of Initial Wait Phase	Timer set of Main Phase	Timer expired [Threshold* = 0]	Set Timer, Send OFFER SERVICE	Not exist: Timer is internal Variables
4	Timer set of Repetition Phase	Timer set of Repetition Phase	Timer expired [run < Threshold*]	Send OFFER SERVICE, Set Timer, run++	Not exist: Timer is internal Variables
5	Timer set of Repetition Phase	Timer set of Repetition Phase	Receive FIND SERVICE	Wait and Send OFFER SERVICE, Reset Timer	Exist: Attacker sends FIND SERVICE
6	Timer set of Repetition Phase	Timer set of Main Phase	Timer expired [run = Threshold*]	Set Timer, Send OFFER SERVICE	Not exist: Timer is internal Variables
7	Timer set of Main Phase	Timer set of Main Phase	Timer expired	Set Timer, Send OFFER SERVICE	Not exist: Timer is internal Variables
8	Timer set of Main Phase	Timer set of Main Phase	Receive FIND SERVICE	Wait and Send OFFER SERVICE, Reset Timer	Exist: Attacker sends FIND SERVICE

* Threshold is equal REPETITION_MAX

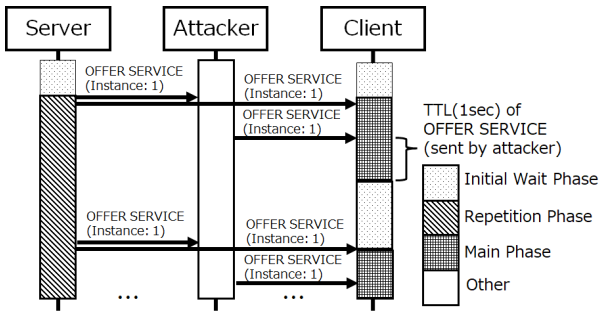

Fig. 4. TTL-Override attack

advantage to the attacker because it extends the time during which the client can use the service. Further, we analyzed state transitions 7 and 10; the attack on these transitions is a TTL-override attack with TTL = 0 (i.e., STOP OFFER SERVICE).

The timer-reset attack at TTL = 0 employs the same method used in the de-association attack reported in a previous study [3], and therefore, the timer-reset attack is considered a generalized version of the DoS attack used in the de-association attack. The timer-reset attack is more powerful when TTL = 0 because it allows the longest DoS time. The analysis results obtained in this study indicate that the method used in the de-association attack reported in a previous study [3] is the most powerful method for performing DoS attacks against clients.

IV. Evaluation

We evaluated the feasibility of the timer-reset attack proposed in Section III (Attack Method) for AUTOSAR SD [9], which specifies the software component specifications of the protocol stack, and vsomeip, which is an implementation of the SOME/IP and SOME/IP-SD protocol stacks. The TTL-override attack was not evaluated in this study because its feasibility in the TTL = 0 case has been reported in a previous study [3].

A. Evaluate Feasibility with AUTOSAR SD

We analyze the specifications to evaluate the effectiveness of the timer-reset attack for the software implemented according to AUTOSAR SD[9].

The timer-reset attack uses the specification that the server resets its internal timer when a FIND SERVICE is received. Section 7.6.2 (Repetition phase for server services) of AUTOSAR SD states “Note: Additionally sent OfferService messages which have been triggered by received FIND SERVICE messages shall have no influence on the counter value of the cyclically OFFER SERVICE messages.” Thus, AUTOSAR SD requires that the specification for resetting the internal timer not be implemented. Therefore, the timer-reset attack is not effective against the software implemented according to AUTOSAR SD.

B. Evaluate Feasibility with vsomeip

We demonstrated the effectiveness of the timer-reset attack with vsomeip 3.3.8[10] using a real device. The evaluation was conducted on a personal computer (PC) with Open-vSwitch 2.17.18[16], installed as the L2 Switch; two PCs (server and client) with vsomeip 3.3.8; and one PC (attacker) with the timer-reset attack implemented. The server is configured to send an OFFER SERVICE every 2000 ms, and the attacker sends a FIND SERVICE every 500 ms. In accordance with Cond. A), the FIND SERVICE was less than half that of the OFFER SERVICE. The client does not send SOME/IP messages to the server to limit the evaluation target to the core of the

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TABLE III
Client state transition table and attacker involvement analysis results

No	Old state	New state	Triger	Action	Attacker influence potential
1	-	Timer set of Initial Wait Phase	Immediate	Set Timer in range (min, max)	Not exist: Triger is Immediate
2	Timer set of Initial Wait Phase	Timer set of Repetition Phase	Timer expired	Send FIND SERVICE, Set Timer, run = 0	Not exist: Timer is internal Variables
3	Timer set of Initial Wait Phase	Service Ready of Main Phase	Receive OFFER SERVICE	Set Timer (TTL)	Exist: Attacker sends OFFER SERVICE
4	Timer set of Initial Wait Phase	Stopped of Main Phase	Timer expired [Threshold = 0]	Send FIND SERVICE	Not exist: Timer is internal variables
5	Timer set of Repetition Phase	Timer set of Repetition Phase	Timer expired [run < Threshold]	Send FIND SERVICE, Set Timer, run++	Not exist: Timer is internal variables
6	Timer set of Repetition Phase	Service Ready of Main Phase	Receive OFFER SERVICE	Set Timer (TTL)	Exist: Attacker sends OFFER SERVICE
7	Timer set of Repetition Phase	Stopped of Main Phase	Receive STOP OFFER SERVICE	None	Exist: Attacker sends STOP OFFER SERVICE
8	Timer set of Repetition Phase	Stopped of Main Phase	Timer expired [run = Threshold]	None	Not exist: Timer is internal variables
9	Service Ready of Main Phase	Service Ready of Main Phase	Receive OFFER SERVICE	Reset Timer (TTL)	Exist: Attacker sends OFFER SERVICE
10	Service Ready of Main Phase	Stopped of Main Phase	Receive STOP OFFER SERVICE	Cancel Timer (TTL)	Exist: Attacker sends STOP OFFER SERVICE
11	Service Ready of Main Phase	Timer set of Initial Wait Phase	Timer (TTL) expired	Set Timer in range (min, max)	Exist: Attacker sends OFFER SERVICE with TTL
12	Stopped of Main Phase	Service Ready of Main Phase	Receive OFFER SERVICE	Set Timer (TTL)	Exist: Attacker sends OFFER SERVICE

* Threshold is equal REPETITION_MAX

proposed attack (that is the feasibility of Cond. A). When evaluating an attack, the server, client, and attacker are started in that order, with the attacker starting after the server has transitioned to the main phase.

An excerpt of the log of SOME/IP messages transmitted and received by the server because of the evaluation is shown in Fig.5. The server transmits an OFFER SERVICE via unicast immediately after receiving the FIND SERVICE. In addition, the server sends a multicast OFFER SERVICE every 2000 ms. This result indicates that vsomeip3.3.8 does not implement a specification for the server to reset its internal timer when it receives the FIND SERVICE. Therefore, the timer-reset attack is not effective against vsomeip3.3.8.

An additional evaluation was performed for demonstrating the effectiveness of the timer-reset attack on vsomeip3.3.8 if the server implements a specification that resets the internal timer when receiving a FIND SERVICE. An additional evaluation was conducted in an environment wherein vsomeip3.3.8 was modified to reset the internal timer when receiving a FIND SERVICE. Fig.6 shows a log of SOME/IP messages sent and received by the server because of the additional evaluation. The results show that the server sends the OFFER SERVICE via unicast after receiving the FIND SERVICE, thereby indicating that the timer-reset attack was successful.

Time[sec]	Source	Destination	Message Type
31.804	Attacker	Multicast	FIND SERVICE
31.805	Server	Attacker	OFFER SERVICE
31.991	Server	Multicast	OFFER SERVICE
32.304	Attacker	Multicast	FIND SERVICE
32.305	Server	Attacker	OFFER SERVICE
32.805	Attacker	Multicast	FIND SERVICE
32.805	Server	Attacker	OFFER SERVICE
...	...	...	...
33.991	Server	Multicast	OFFER SERVICE
34.305	Attacker	Multicast	FIND SERVICE
34.306	Server	Attacker	OFFER SERVICE

Fig. 5. Evaluation result with vsomeip

C. Attack Countermeasures

All conditions required to execute a timer-reset attack are attacker-dependent, and therefore, servers cannot implement countermeasures if they comply with the SOME/IP-SD specifications. Therefore, it is possible to counter a time-reset attack by not supporting either of the following specifications:

Spec. A) Send OFFER SERVICE by Unicast when FIND SERVICE is received

Spec. B) Reset the timer that measures the transmission

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Time[sec]	Source	Destination	Message Type
34.634	Attacker	Multicast	FIND SERVICE
34.635	Server	Attacker	OFFER SERVICE
35.134	Attacker	Multicast	FIND SERVICE
35.135	Server	Attacker	OFFER SERVICE
35.635	Attacker	Multicast	FIND SERVICE
35.635	Server	Attacker	OFFER SERVICE
36.635	Attacker	Multicast	FIND SERVICE
36.636	Server	Attacker	OFFER SERVICE
37.135	Attacker	Multicast	FIND SERVICE

Fig. 6. Additional evaluation result with vsomeip

cycle of the OFFER SERVICE when the FIND SERVICE is received

If Spec. A) is not supported (i.e., OFFER SERVICE is sent only via multicast), an attacker may send a FIND SERVICE and conduct a DoS attack that increases the communication load. If Spec. B) is not supported, the communication load will also increase. However, the processing does not change depending on requests from external devices, and therefore, this is considered a more effective countermeasure instead of not supporting Spec. A). The effectiveness of not supporting Spec. B) as a countermeasure against the timer-reset attack is shown in Subsection IV-B (feasibility evaluation with vsomeip) and Figure 5.

The TTL-override attack can be considered in the same manner as the timer-reset attack because Cond. C) is an attacker-dependent condition, and it effectively prevents the need to satisfy Cond. D). However, this countermeasure is effective only for TTL-override attacks with TTL > 0; it is not effective against TTL-override attacks with TTL $= 0$. Therefore, it is necessary to consider other countermeasures against TTL-override attacks such as authentication and IDS.

V. Conclusion and Future Work

We analyzed the possibility of attacks on disabled service provisioning and service usage through an exhaustive analysis of the state-transition model of SOME/IP-SD. The analysis showed that an attack on disabled service provisions (timer-reset attack), which has not been reported in previous studies, is possible. We demonstrated the feasibility of the proposed attack by analyzing the AUTOSAR SD specifications and evaluating the attack on real devices using vsomeip. The evaluation results indicated that the proposed attack was not effective for these software packages because it did not implement the specification to reset the internal timer upon receiving a SOME/IP-SD FIND SERVICE used in the attack. An additional evaluation using vsomeip showed that the proposed attack was effective when the specification for resetting the internal timer was implemented. Based on the evaluation results, an example of a countermeasure against a time-reset attack was presented.

We demonstrated the possibility of attacks on the disabled service provision and countermeasures against such attacks; we did not show any effective countermeasures besides authentication against attacks to deny service use. In future studies, we plan to discuss additional countermeasures against DoS attacks, including specific authentication and other similar methods.

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